Roadkill Hotspots and Policy Advocacy (RHPA): A Multi-Layered AI and IoT Approach for Real-Time Wildlife Preservation

Lokesh Patra & Indigibilli Harshit
Computer Science and Engineering
Sri Sri University
Cuttack, India
lokesh0patra@gmail.com | harshitindigibilli@gmail.com

***Abstract*—Roadkill incidents represent a critical intersection of ecological loss and vehicular safety hazards. With rapid urbanization, the frequency of wildlife-vehicle collisions has escalated, resulting in significant biodiversity loss and danger to commuters. This research proposes a novel, three-dimensional (3D) solution titled "Roadkill Hotspots and Policy Advocacy" (RHPA). Unlike traditional radar-based systems, RHPA integrates Computer Vision (YOLOv5xu), Ensemble Deep Learning (DenseNet121 + EfficientNet-B3), and Internet of Things (IoT) to create a preventative and responsive ecosystem. The system achieves an overall accuracy of ≈95%, employing YOLOv5xu for real-time detection and a CNN ensemble for classifying subjects as "Alive" (High Risk) or "Dead" (Low Risk). Furthermore, the system integrates an ESP32-based actuation layer for real-time driver warnings and a policy-level dashboard for identifying blackspots. Pilot testing indicates a potential 36% reduction in roadkill incidents.**

***Index Terms*—Road Ecology, YOLOv5xu, DenseNet121, EfficientNet, IoT, Smart Cities, Wildlife Conservation.**

I. Introduction

The expansion of road networks acts as a double-edged sword; while it facilitates economic growth, it fragments natural habitats, forcing wildlife to navigate hazardous asphalt terrains. Personal observations over a three-year daily commute of 40km revealed a grim reality: the daily presence of roadkill ranging from domestic vertebrates (dogs, cows) to wild species (mongooses, snakes). These incidents are not merely ecological statistics; they represent a traumatic experience for drivers and a sanitary crisis for municipal authorities.

Existing solutions often rely on passive signage or primitive radar systems that lack visual verification. This paper introduces a **3D AI-based solution**:

- **Dimension 1 (Vision):** Real-time detection and tracking.
- **Dimension 2 (Classification):** Distinguishing between live threats and carcasses to automate distinct responses.
- **Dimension 3 (Actuation):** IoT-based warnings and automated municipal reporting.

II. Literature Review

The foundation of this research is built upon analyzing gaps in existing road ecology frameworks.

A. Acoustic and Radar Systems

Ahn and Bae proposed a "Sound Warning System for Prevention of Roadkill" utilizing radar to detect movement and emit ultrasonic frequencies [1]. While their approach successfully deters some animals, radar systems suffer from a high false-positive rate (triggering for inanimate objects) and cannot distinguish between a living animal crossing the road and an already deceased one. Our proposed RHPA system overcomes this by replacing radar with Computer Vision (YOLOv5xu) for granular object classification.

B. Road Ecology and Traffic Theory

Research by van der Ree et al. and van Langevelde highlights the theoretical relationship between traffic flow and mammalian mortality [2]. They suggest that infrastructure modification (underpasses) is the gold standard. However, such infrastructure is expensive and slow to implement. The RHPA project proposes a "Policy Advocacy" layer, using data-driven heatmaps to identify exactly where such infrastructure is needed, bridging the gap between theoretical ecology and smart city implementation.

III. System Architecture

The proposed methodology follows a three-tiered architecture combining Edge AI and Cloud Analytics.

A. D1: Object Detection (YOLOv5xu)

The first dimension utilizes the YOLOv5xu architecture. This variant is optimized for identifying small-to-medium objects (animals) in complex road backgrounds.

- **Input:** Real-time feeds from dashcams or roadside surveillance.
- **Function:** Bounding box regression and optical flow-based speed tracking.

The loss function considers the Intersection over Union (IoU) to minimize localization errors:

$$L_{IoU} = 1 - IoU + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v \tag{1}$$

Where ρ is the Euclidean distance between center points, and αv represents the aspect ratio consistency.

B. D2: Classification (Ensemble Learning)

Once an animal is detected, the Region of Interest (ROI) is passed to a classification module to determine the state: *Alive* vs. *Dead*. We employ an ensemble of two distinct CNN architectures to maximize reliability.

1) DenseNet121: Utilizes dense connectivity to mitigate the vanishing gradient problem, ensuring that feature maps are preserved throughout the network. This is crucial for detecting subtle features like breathing or posture.

2) EfficientNet-B3: Employs compound scaling to balance depth, width, and resolution, providing high accuracy with lower computational cost.

3) Ensemble Logic: The final classification probability P_{final} is derived using a weighted average:

$$P_{final} = w_1 \cdot P_{DenseNet} + w_2 \cdot P_{EfficientNet} \quad (2)$$

Where $w_1 + w_2 = 1$. Based on validation metrics, we assign $w_1 = 0.5$ and $w_2 = 0.5$.

C. D3: IoT and Actuation

The hardware layer is controlled by an ESP32 microcontroller.

- **Sensors:** Ultrasonic sensors for proximity verification; DHT sensors for environmental correlation (temp/humidity).
- **Actuators:** Smart signage (LED Matrices) and in-car buzzers (via TinyML integration).

IV. Proposed Algorithm & Logic Flow

The core logic distinguishes between preventive safety (Alive) and corrective action (Dead).

```
Initialize Camera, ESP32, Models
while System is Active do
    Frame ← CaptureFrame()
    Objects ← YOLOv5xu.Detect(Frame)
    for each Object in Objects do
        if Object is Animal then
            State ← EnsembleModel.Classify(Object)
            if State == ALIVE then
                Trigger: High Risk Alert
                Action: Activate Buzzer (Beep)
                Action: Auto-decelerate Vehicle (No sudden break)
                Log: Update Dynamic Danger Map
            else if State == DEAD then
                Trigger: Low Risk / Obstacle Alert
                Action: Notify Driver (Avoidance)
                Action: Send GPS to Municipal Corp (Cleanup)
                Log: Update Roadkill Hotspot Database
            end if
        end if
    end for
end while
```

Fig. 1. RHPA Decision Logic

V. Implementation and Results

A. Experimental Setup

The models were trained on a custom dataset comprising various vertebrates (cows, dogs, snakes, etc.) in diverse lighting conditions. The system was pilot-tested across 4 locations over 1.5 months.

B. Performance Metrics

The models were evaluated based on Precision, Recall, and F1-Score.

TABLE I
Model Performance Comparison

Model	Task	Accuracy
YOLOv5xu	Object Detection	93.0%
DenseNet121	Classification (State)	98.0%
EfficientNet-B3	Classification (State)	97.0%
CNN Ensemble	**Classification (Combined)**	**97.5%**

The overall system accuracy (Detection + Classification pipeline) is approximately **95%**.

C. Impact Analysis

During the pilot phase, the system's interventions (smart signage warnings and driver alerts) resulted in an estimated **36% reduction** in roadkill incidents compared to historical data for the same route.

VI. Policy Advocacy and Heatmaps

Beyond immediate prevention, this project serves a long-term legislative goal. By logging the GPS coordinates of every "Dead" classification, we generate high-resolution Heatmaps.

Application for Authorities:

1) **Hotspot Identification:** Pinpointing specific road segments (Blackspots) where animals frequently cross.
2) **Infrastructure Planning:** Data-backed recommendations for constructing underpasses or overpasses.
3) **Resource Allocation:** Optimizing the routes for municipal cleanup crews based on real-time notifications.

VII. Conclusion

The RHPA project successfully bridges the gap between advanced technology and environmental ethics. By moving beyond simple detection to complex state classification (Alive vs. Dead), the system offers tailored responses that ensure passenger safety while actively preserving wildlife. The integration of this data into municipal policy frameworks ensures that the solution is not just reactive, but curative for the long-term health of our road ecosystems. Future work will focus on expanding the dataset to include nocturnal species and integrating thermal imaging for night-time detection.

REFERENCES

[1] I. S. Ahn and M. J. Bae, "A Study on the Sound Warning System for Prevention of Roadkill Accident," *International Journal of Engineering Research and Technology*, vol. 11, no. 10, pp. 1605-1618, 2018.

[2] R. van der Ree, D. J. Smith, and C. Grilo, *Handbook of Road Ecology*. Chichester, UK: Wiley, 2015.

[3] A. Torres et al., "Assessing large-scale wildlife responses to human infrastructure development," *Proc. Nat. Acad. Sci.*, vol. 113, no. 30, pp. 8472–8477, 2016.

[4] F. van Langevelde and C. F. Jaarsma, "Using traffic flow theory to model traffic mortality in mammals," *Landscape Ecol*, vol. 19, no. 8, pp. 895–907, 2005.

[more to be added*]